

DEBRAJ BOSE

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RESEARCH INTERESTS

My research is in statistical genetics, with a focus on developing statistical methods for gene-level inference and causal gene prioritization in human genetic studies. I am also broadly interested in applied statistical modeling, clinical trial design and analysis, and machine learning, with an emphasis on principled inference and real-world data applications.

EDUCATION

University of Michigan, Ann Arbor 2019 – 2026 (expected)

PhD in Biostatistics, GPA: 4.0/4.0

Thesis: Statistical Methods for Gene-Level Inference and Causal Gene Prioritization in Human Genetic Studies

Co-advised by: Prof. Michael Boehnke and Prof. Xiaoquan (William) Wen

Indian Statistical Institute, Delhi and Kolkata, India

M.Stat. (Biostatistics Specialization), Overall Score: 79.5% 2017 – 2019

St. Xavier's College, Kolkata, India

B.Sc. (Hons) in Statistics, GPA: 8.9/10 2014 – 2017

PUBLICATIONS

- **Bose D**, Fuchsberger C, Boehnke M. Rare-variant association studies: When are aggregation tests more powerful than single-variant tests? *American Journal of Human Genetics* 112(8), 1948-1961 (2025).
- Yin X, Li J, **Bose D**, ... , Morrison J. Assessing the potential causal effects of 1099 plasma metabolites on 2099 binary disease endpoints. *Nature Communications* 16, 3039 (2025).
- Yin X, **Bose D**, Kwon A, ... , Wen X. Integrating transcriptomics, metabolomics, and GWAS helps reveal molecular mechanisms for metabolite levels and disease risk. *American Journal of Human Genetics* 109(10), 1727–1741 (2022).
- Yin X, Chan LS, **Bose D**, ... , Boehnke M. Genome-wide association studies of metabolites in Finnish men identify disease-relevant loci. *Nature Communications* 13, 1644 (2022).
- Ray D, Salvatore M, Bhattacharyya R, ... , **Bose D**, ... , Mukherjee B. Predictions, role of interventions and effects of a historic national lockdown in India's response to the COVID-19 pandemic: data science call to arms. *Harvard Data Science Review*, Suppl. 1 (2020).

POSTERS AND TALKS

- Poster, *American Society of Human Genetics Annual Meeting*, Boston, October 2025.
- Poster, *Eastern North American Region Spring Meeting*, New Orleans, March 2025.
- Contributed talk, *Joint Statistical Meetings*, Portland, August 2024.
- Poster, *American Society of Human Genetics Annual Meeting*, Washington DC, November 2023.
- Poster, *International Chinese Statistical Association Applied Statistics Symposium*, Ann Arbor, June 2023.
- Invited talk, *Graduate Student Working Group, Dept. of Biostatistics, University of Michigan*, Ann Arbor, February 2023.
- Poster, *American Society of Human Genetics Annual Meeting*, Los Angeles, October 2022.

- Poster, *American Society of Human Genetics Annual Meeting*, Virtual Conference, October 2020.
- Poster, *Indian Science Congress*, Mysore, India, January 2016.

RESEARCH EXPERIENCE

University of Michigan, Ann Arbor

Graduate Research Assistant

Sep 2019 – Present

- Developed method to estimate gene-level posterior probabilities of causality for a given complex disease/phenotype by integrating Genome-Wide Association Studies (GWAS), molecular Quantitative Trait Loci (QTLs), and gene-level features in an **empirical Bayes framework** using an **EM algorithm**. (Manuscript in progress)
- Provided guidelines for practitioners by performing analytic calculations and extensive simulations for rare-variant association analyses on **UK Biobank exome data** ($n \sim 408K$, number of rare variants $\sim 6.7M$).
- Collaborated on the **METSIM study** to analyze high-dimensional metabolomics data, using gene-based aggregation methods to detect gene-metabolite associations for $\sim 1.4K$ metabolites.

Indian Statistical Institute, Kolkata, India

Master's Project

Aug 2018 – May 2019

Advisor: Prof. Subir K. Bhandari

- Proposed optimal histogram construction in one and multiple dimensions by (1) using **k-means clustering and Voronoi tessellations**, and (2) minimization of mean integrated squared error.
- Developed a novel supervised classification method based on **optimal histogram construction** and demonstrated through simulations across multiple settings that these classifiers achieve performance comparable to optimal Bayes classifier.
- Strengthened expertise in statistical learning theory, density estimation, and algorithmic implementation.

Reliance Industries Limited, Navi Mumbai, India

Summer Intern

May 2018 – July 2018

- Built NLP-based pipelines in **Python** to extract, summarize, and classify information from quarterly financial reports of major oil and gas competitors, and also automated financial table generation and report summarization, reducing manual analysis time and improving data consistency.
- Collaborated with business analysts to translate unstructured text into structured insights for strategic decision-making.

AWARDS AND HONORS

· Rackham Graduate School Conference Travel Grants	2022–2025
· Sabyasachi Roy Gold Medal award for best M.Stat. project	2019
· Book Grant for exemplary performance in M.Stat. examinations	2018–2019
· INSPIRE scholarship from Government of India	2014–2017
· Bronze medals representing North Kolkata table tennis team in West Bengal State Championships	2009, 2011

LANGUAGE, PROGRAMMING AND STATISTICAL SKILLS

Programming languages and frameworks: R, LaTeX, Linux (expert); Python, C++, Git (practitioner).

Relevant coursework: Probability, Statistical Inference, Linear Regression, Multivariate Analysis, Longitudinal Data Analysis, Statistical Computing, Clinical Trials, Statistical Genetics, Survival Analysis, Pattern Recognition.

Languages: English, Bengali, Hindi.

Summary of statistical and computational skills:

- Strong background in analyzing and performing simulations with **large-scale genetic and metabolomic datasets** in high-performance **Linux** computing environments.

- **Statistical modeling and inference in R**, including longitudinal, survival, and clinical trial data analysis.
- Developed visualizations and interactive dashboards using **R Shiny, ggplot2, plotly, and flexdashboard**.

SERVICE AND LEADERSHIP

Statistics in the Community (STATCOM), University of Michigan

Communications Chair (2024–2025), Member (2023–2025)

- Led and contributed to pro bono data science projects for nonprofit and community organizations.
- Developed interactive **flexdashboard** visualizations for the Oakland Livingston Human Service Agency to understand needs of people experiencing hardship and poverty in Southeast Michigan.
- Built an **R Shiny dashboard** to visualize country-level and state-level visitor demographics and geographic trends for the Ann Arbor Hands-On Museum.
- Led a team project to analyze and summarize door-to-door survey data in two impoverished neighborhoods for The People's Action, producing a research report to support community advocacy efforts.
- Led a team project to design summary analyses and visualizations, including text summaries and word clouds, for program evaluation surveys for the Center for Success literacy initiative.

Vice-President (2020–2022) and A-Team Member (2019–2025) of University of Michigan Table Tennis Team

- Won team and individual awards for University of Michigan at the Divisional and Regional tournaments.

REFERENCES

Michael Boehnke, PhD

Richard G. Cornell Distinguished University Professor of Biostatistics

University of Michigan, Ann Arbor

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Xiaoquan (William) Wen, PhD

Professor, Biostatistics

University of Michigan, Ann Arbor

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Christian Fuchsberger, PhD

Research Group Leader

Institute for Biomedicine, Eurac Research, Bolzano, Italy

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Xianyong Yin, MD

Professor, Epidemiology

Nanjing Medical University, Nanjing, Jiangsu, China

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